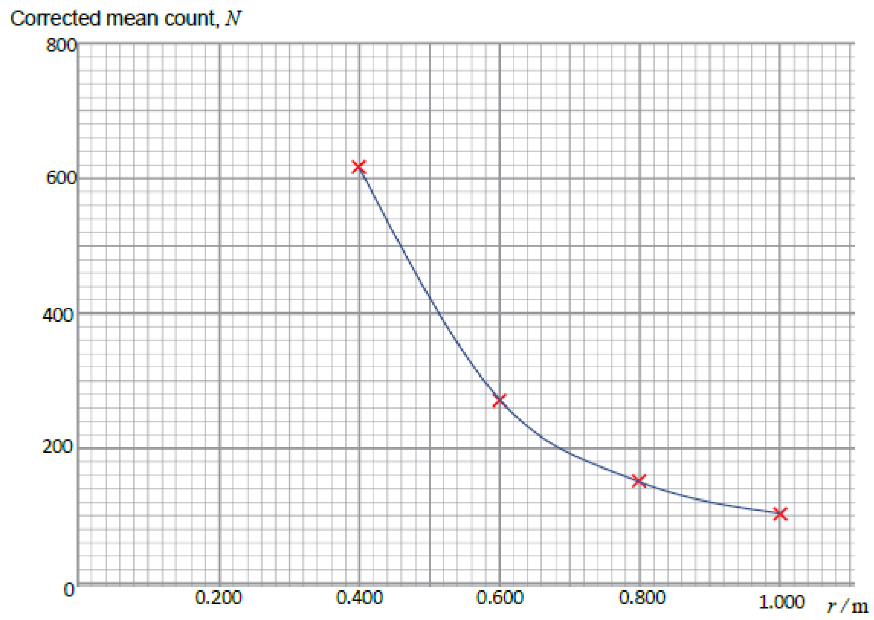
<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>5</td><td>(a)</td><td>(i)</td><td>Absorb alpha and beta or only allow gamma through. Accept absorb beta only [due to 40 cm gap]</td><td></td><td></td><td>1</td><td>1</td><td></td><td>1</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Background radiation (1)
To check for consistency / check for variations (1)
Don’t accept for a mean</td><td></td><td></td><td>2</td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td>(b)</td><td></td><td>
<table>
<tr><th rowspan="2">Distance, r / m</th><th colspan="4">Count</th><th rowspan="2">Uncertainty in mean count</th></tr>
<tr><th>First reading</th><th>Second reading</th><th>Third reading</th><th>Mean count</th></tr>
<tr><td>Without source</td><td>19</td><td>22</td><td>25</td><td>22</td><td>3</td></tr>
<tr><td>1.000</td><td>110</td><td>130</td><td>136</td><td>125</td><td>13</td></tr>
<tr><td>0.800</td><td>195</td><td>165</td><td>178</td><td>179</td><td>15 or 14 or 16</td></tr>
<tr><td>0.600</td><td>270</td><td>316</td><td>300</td><td>295</td><td>23 or 21 or 22 or 24 or 25</td></tr>
<tr><td>0.400</td><td>661</td><td>604</td><td>651</td><td>639</td><td>29</td></tr>
<tr><td>Without source</td><td>20</td><td>20</td><td>25</td><td>22</td><td>3</td></tr>
</table>
1 mark per column.</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td>2</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>
<table>
<tr><th>Distance, r /m</th><th>Corrected mean count, N</th><th>Uncertainty in corrected mean count</th></tr>
<tr><td>1.000</td><td>103</td><td>16</td></tr>
<tr><td>0.800</td><td>157</td><td>18</td></tr>
<tr><td>0.600</td><td>273</td><td>26</td></tr>
<tr><td>0.400</td><td>617</td><td>32</td></tr>
</table>
1 mark per column ecf on previous table</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td>2</td></tr>
</table>

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
		(ii)	 All 4 points correct (2) **ecf** 3 points correct (1) One smooth curve passing through all points ± ½ small square (1)		3		3	3	3
		(iii)	Error bars in the corrected mean count are large enough to be shown on the graph [approx. 2 units min], but the uncertainty in r is too small [less than one unit]			1	1		1

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		$\ln$ 119 = 4.78 **or** $\ln$ 87 = 4.47 (1) $\ln$ 103 = 4.63 **or** the other extreme (1) 4.78 - 4.63 **or** 4.66 - 4.47 = [approx. 0.15] or $\frac{0.31}{2}$ (1) **No ecf** allowed (since 103 and 16 are given) **Award 1 mark** for $\frac{16}{103} = 0.15.\ 0.155, 0.16$ **Award 2 marks** for statement absolute uncertainty in log = fractional uncertainty in count		3		3	3	3
	(e)	(i)	Taking logs: $\ln(N) = -n\ln(r) + \ln(k)$ (1) (accept $\ln I$ in lieu of $\ln N$) Plotting $\ln(N)$ on the y-axis against $\ln(r)$ on the x-axis or comparison with $y = mx + c$ (1) Gradient = $-n$ (1)			3	3	3	3
		(ii)	Minimum gradient = $\frac{3.94-7.00}{0.50-(-1.30)}$ = $[-]1.69$ ± 0.03 (1) Maximum gradient = $\frac{3.40-7.00}{0.50-(-1.17)}$ = $[-]2.16$ ± 0.03 (1) Mean gradient = $[-]1.93$ ± 0.03 (1) **ecf** Uncertainty = $\frac{1}{2}(-1.69-(-2.16))$ = 0.20 - 0.28 (1) **ecf** accept 0.2 No sig fig penalty		4		4	4	4
			Question 5 total	**0**	**14**	**7**	**21**	**17**	**21**